
Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

Abdulrahman Alswaidan

Robert Frederick Smith School of Chemical and Biomolecular Engineering
Cornell University, Ithaca, NY 14850
aa2725@cornell.edu

Jeffrey D. Varner*

Robert Frederick Smith School of Chemical and Biomolecular Engineering
Cornell University, Ithaca, NY 14850
jdv27@cornell.edu

Abstract

Synthetic multi-asset equity paths must preserve both the per-asset return distributions and their relationship with the market. A common construction inserts a full-return generator draw directly into a single-index model, but this double counts the market variance and distorts the composed marginal. We derived a closed-form scale correction for this problem and evaluated it on 423 non-market assets in a 424-asset US equity and exchange-traded-fund universe. The correction set each synthetic asset's second moment to its empirical target, recovered the calibrated factor loading with low error, and preserved heavy-tailed marginals, all without a separate fit to the regression residuals. Residual models fitted directly to each asset passed the marginal goodness-of-fit tests more often, but only at the cost of a per-asset calibration stage. On 416 complete asset histories held out from 2025, the correction raised the cross-ticker pass rate on the same marginal tests, brought the median ratio of synthetic to empirical variance close to one, and produced one-day 99% Value-at-Risk exceedances comparable to the residual-fit counterpart, though above the nominal rate. The per-asset input was a hidden Markov model with Laplace-quantile states and Student-t emissions, whose jump-duration variant used Poisson episodes to lengthen visits to the tail states. On a decade of daily SPY data, this variant lowered the absolute-return autocorrelation error relative to its no-jump counterpart while keeping strong distributional fit, and we selected it as the marginal component on this tradeoff, not because it dominated every baseline. The multi-asset construction still omits cross-asset residual covariance, so independent residuals can overstate diversification gains under daily rebalancing. Code and the per-asset fitted marginals are released with the paper.

Keywords: hidden Markov model; jump-duration mechanism; single-index model; synthetic financial data; variance correction; Value-at-Risk coverage check; volatility clustering; heavy-tailed marginals.

1 Introduction

Synthetic financial data often fail to reproduce the statistical properties of real returns. These paths are used to stress-test risk models against unobserved market scenarios, validate portfolio optimization

*Corresponding author.

algorithms, and Monte Carlo backtest allocators across many independent scenarios [1, 2]. Empirical equity excess growth rates exhibit three statistical regularities a generative model must reproduce: a heavy-tailed marginal distribution, weak autocorrelation of raw growth rates, and persistent volatility clustering, in which large changes are followed by large changes and small changes by small changes [3–5]. All three patterns are visible in SPY daily excess growth rates from 2014–2024 (Fig. 1). The empirical density is sharply concentrated near zero but places substantially more probability in the tails than a Gaussian model, as shown by the density and quantile-quantile comparisons (Fig. 1a and Fig. 1b). Raw growth rates have little serial correlation (Fig. 1c), whereas the positive, slowly decaying autocorrelation of their absolute values (Fig. 1d) shows that their magnitudes remain temporally dependent. These regularities, called *stylized facts*, demonstrate why matching the marginal distribution alone is insufficient: synthetic financial data must also preserve the distinct dependence structures of return levels and return magnitudes.

Existing generators typically reproduce some of these regularities but not others. The generalized autoregressive conditional heteroskedasticity (GARCH) family [6, 7] captures volatility clustering but requires fat-tailed innovations to reproduce heavy tails. Stochastic-volatility and jump-diffusion models [8–10] produce heavy tails, but jump effects do not persist. Deep generative models [11–13] can learn complex marginals but often miss the absolute-return autocorrelation associated with volatility clustering. Standard hidden Markov models (HMMs) [14] provide a regime-switching structure but return to central regimes too quickly to dwell in tail states long enough to reproduce the empirical clustering profile [15, 16]. We therefore sought a model that retained distributional fit, reduced absolute-return autocorrelation error, and could be fitted separately across a large asset universe.

In this study, we developed a multi-asset equity simulation framework that combines separately fitted hidden Markov marginals with a variance-corrected single-index composition. Each marginal model used a Laplace-quantile state partition, Student- t emissions, and direct transition counting, while the jump-duration variant (HMM-WJ) added Poisson-distributed tail-state episodes. On a decade of SPY data, HMM-WJ had a 97.6% in-sample KS pass rate and reduced ACF-MAE from 0.059 to 0.052 relative to the no-jump variant, although GARCH reached 0.031. This tradeoff motivated our use of the hidden Markov family as the per-asset marginal rather than establishing HMM-WJ as uniformly superior. For multi-asset composition, we showed that using a full per-asset generator draw as the residual in a single-index model adds the market variance twice, and we derived a closed-form scale correction that preserved target variance and recovered the calibrated factor loading across a 424-asset universe. When the 2014–2024 fits were frozen and evaluated on 416 complete asset histories from 2025, the correction raised the mean cross-ticker KS pass rate from 78.8% to 84.1% and reduced the median variance ratio from 1.29 to 0.97. Independent residuals nevertheless exaggerated daily cross-sectional dispersion and could overstate diversification gains in daily-rebalanced portfolios; setting a location-shift adjustment to a long-run prior reduced this return-level bias but did not restore the missing cross-asset dependence. By clarifying how marginal fidelity, variance preservation, and dependence interact, this framework can support the generation of more reliable multi-asset synthetic financial data for large-scale stress testing and portfolio evaluation.

2 Related Work

Models of equity-market stylized facts follow either a bottom-up or top-down approach. Bottom-up agent-based models [17–22] derive return dynamics from simulated interactions among heterogeneous agents. They are demanding to calibrate and expensive to simulate at scale. Top-down models instead fit statistical structures directly to observed growth-rate series. Top-down generators trace to Bachelier’s diffusion model, later formalized as geometric Brownian motion and used by Black-Scholes-Merton for option pricing [23–26]; these diffusion baselines are tractable but fail on the empirical regularities that drive risk and pricing. Mandelbrot [3] showed returns have heavier tails than the Gaussian admits; Fama [27] reviewed evidence that raw returns have near-zero autocorrelation; and Schwert [28] documented persistence in monthly absolute residuals, later codified by Cont [4] as volatility clustering. The autoregressive conditional heteroskedasticity (ARCH) and generalized ARCH (GARCH) models of Engle and Bollerslev [6, 7] reproduced clustering but required a heavy-tailed noise term to produce heavy-tailed marginals. Neither model includes discrete regimes or sudden jumps. Stochastic-volatility models [29, 8] make volatility a latent process to capture clustering, but estimating that process requires filtering or Markov chain Monte Carlo (MCMC)

methods [30]; jump-diffusion models [9, 10] inject Poisson events into the price path, producing heavy tails by construction, but volatility does not persist after the jump.

Hidden Markov models embed regime and jump behavior directly in a latent discrete chain, following Hamilton’s [31] two-state regime-switching formulation, later applied to volatility forecasting [32] and equity trading [33]. Ryden et al. [15] found that Gaussian-emission HMMs matched heavy tails and the absence of return autocorrelation but did not reproduce volatility clustering; Bulla and Bulla [16] modeled clustering with hidden semi-Markov models (HSMMs) that replaced the geometric dwell-time distribution of every state with a parametric (negative-binomial) alternative. Both rely on the Baum-Welch expectation-maximization (EM) algorithm [34], which is iterative and sensitive to initialization. Deep generative alternatives learn the marginal directly but routinely miss the volatility-clustering signature: Takahashi et al. [11] found that generative adversarial network (GAN) architectures failed to reproduce the absolute-return autocorrelation function, and Kwon and Lee [13] reported the same failure for TimeGAN [12]. Recent surveys [1, 2, 5] agree that stylized-facts reproduction remains the primary quality criterion. Our marginal component also used a latent-state chain, but it differed from the HSMM. It kept the empirically counted transition matrix for the central states and added an event-driven override that forces the chain into the tail states for a Poisson-distributed duration on a subset of time steps.

Multi-asset composition adds cross-asset dependence on top of a univariate generator. Mixture hidden Markov models [35] share latent state across assets but are costly to estimate in high dimensions. Copula approaches [36–38] separate marginals from cross-asset dependence but require choosing a copula family and, in their standard static form, treat successive time slices as independent. Factor models instead reduce a firm’s growth rate to one or more shared factors [39, 40]. The single-index model (SIM) of Sharpe [41] uses one shared factor and decomposes each asset’s growth rate into a market term and an idiosyncratic residual. This structure can use a single-factor market generator directly. We derived a closed-form correction that preserves the generator variance while recovering the calibrated market loading. To evaluate the resulting paths, we combined goodness-of-fit tests for the return distribution [42–44] with autocorrelation diagnostics for temporal structure [4]. These tests are necessary but not sufficient: a synthetic dataset can pass them yet still mislead when fed into the task it was generated for. Booth and Fama [45] described one such failure: a daily-rebalanced allocator on paths whose residuals are drawn independently across assets extracts a structural *diversification return* that is muted in real markets, where common sector and style factors correlate residuals across assets. We adopted the classical distributional metrics together with a Value-at-Risk coverage check scored by Kupiec’s unconditional-coverage test, and addressed this diversification-return bias with a location-shift correction.

3 Methods

We fitted a marginal generator for each asset and then combined the generated paths with a variance-corrected single-index model. The dataset contained 424 United States-listed equities and exchange-traded funds observed from 2014 through 2024. The single-asset analysis focused on the SPDR S&P 500 ETF (SPY). We used 2,766 observations from January 3, 2014, through December 31, 2024, for fitting and the next 249 trading days of 2025 for out-of-sample testing. Of the 423 non-market assets in the training universe, 416 had complete 2025 histories and entered the multi-asset holdout evaluation. The data included daily open, high, low, and close prices and volume-weighted average prices. We computed the excess growth rate $G_{i,j}$ for ticker i between trading days $j - 1$ and j from the equity price series $P_{i,j}$ as

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f, \quad (1)$$

where $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate derived from Treasury STRIPS (Separate Trading of Registered Interest and Principal of Securities) bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

3.1 Per-asset hidden Markov marginal generator

We defined the hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ) as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [14] (Figure 2). The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represents a

market regime, and the observation at step t is the excess growth rate G_t of Eq. (1), drawn from the continuous observation space $\mathcal{O} \subset \mathbb{R}$ (we suppress the ticker index i when only one asset is in play). The transition matrix $\mathbf{T}$ governs regime-to-regime transitions, $\mathbf{E}$ is the state-conditional emission model, and $\bar{\pi}$ is the stationary distribution. We defined the states using quantile boundaries from a fitted Laplace cumulative distribution: $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$, where $F_L^{-1}(\cdot; \mu_L, b_L)$ is the inverse cumulative distribution function of the Laplace distribution with location μ_L and scale b_L . We estimated both parameters from the in-sample growth-rate history by maximum likelihood and fixed the endpoints at $Q_0 = -\infty$ and $Q_N = +\infty$. We assigned an observation to state k when $Q_{k-1} < G_t \leq Q_k$. We chose the Laplace distribution because its sharp central peak follows the concentration of small price movements in the empirical marginal density (Fig. 1(a)) [46, 47], while its closed-form quantile function avoids iterative optimization across the 424-ticker universe [48].

The within-state emission followed a location-scale Student- t distribution:

$$G_t \mid S_t = k \sim \mu_k + \sigma_k \cdot t_\nu, \quad \nu = 5, \quad (2)$$

where t_ν denotes a standard Student- t random variable with ν degrees of freedom. For observations assigned to state k , μ_k is the sample mean and σ_k is the sample standard deviation. A standard t_ν has variance $\nu/(\nu - 2)$, so σ_k is a scale parameter rather than the emission standard deviation. At $\nu = 5$, that standard deviation is $\sigma_k \sqrt{5/3}$. We selected $\nu = 5$ from a sensitivity sweep ($\nu = \infty$ recovers the Normal baseline; explored range in Table 1). At $\nu = 5$, simulated kurtosis was 7.6 against the observed 7.7, compared with 5.5 under Normal emissions; the corresponding KS pass rates were 97.6% and 97.0%. The full comparison against a Gaussian-emission HSMM baseline of Bulla and Bulla [16] is given in Table S3 of Sec. S5.

With the state space and emission family fixed, we estimated the transition matrix $\mathbf{T}$ by counting observed transitions. This avoided the initialization sensitivity of iterative expectation-maximization [48]. Direct counting assigns low probabilities to moves from central states into tail states, so the resulting model loses volatility clustering too quickly [16]. We added a jump mechanism to extend tail-state visits. Its two hyperparameters were the jump probability ϵ and the mean episode length λ . Tail states were defined as $\mathcal{S}_- = \{1, \dots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \dots, N\}$, where N_{tail} was user-configurable (value in Table 1). When a jump occurred, we sampled its length as $K \sim \text{Poisson}(\lambda)$ [49]. Because the Poisson distribution includes zero, $K = 0$ produced no forced tail steps and the chain made an ordinary transition. Thus, λ set the mean episode length, not a minimum dwell time. During the K forced steps, the jump mechanism replaced the standard transitions with draws from a tail set. It selected $\mathcal{S}_-$ with probability p_{neg} and $\mathcal{S}_+$ otherwise (values in Table 1). The negative-tail bias reproduced the gain/loss asymmetry shown in Figure 2. The empirical transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence, with natural residence times of 1–2 steps in every state including the tails (Fig. S5 in Online Appendix S4). The stationary distribution $\bar{\pi}$ satisfies $\bar{\pi} = \bar{\pi} \mathbf{T}$ and was estimated by matrix powering ($\mathbf{T}^{50}$) [50, 51]. We drew the initial state from $S_0 \sim \bar{\pi}$. Because $\bar{\pi}$ is the invariant distribution of $\mathbf{T}$ alone, it only approximates the invariant distribution of the jump-augmented process, whose state also includes jump status and remaining episode length; this introduces a small, unquantified deviation from exact stationarity. Algorithm 1 gives the full simulator. At each step, the chain either made a standard transition with probability $1 - \epsilon$ or entered a Poisson jump episode. A jump forced K draws from a tail set before standard transitions resumed. The model sampled $G_t \sim \mathbf{E}_{S_t}$ after each transition.

We calibrated the jump probability ϵ and mean episode length λ . The mechanism followed the jump-event tradition of [9, 10], but replaced instantaneous jumps with an HSMM-style duration [16]. HMM-WJ is a discrete-time state model: the override extends visits to tail states but adds no continuous-time diffusion component. We selected (ϵ, λ) by grid search against two features of the SPY history in Fig. 1: volatility clustering and kurtosis [3, 4, 15]. The first term of the objective measured the squared error between the observed and simulated ACF of absolute excess growth rates through lag L . The second penalized the difference in global kurtosis. Most observed ACF variation occurred within the calibration window listed in Table 1; the final comparison in Fig. 3 extends to lag 252:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L (\text{ACF}_{\text{obs}}(\tau) - \overline{\text{ACF}}_{\text{sim}}(\tau))^2 + w_\kappa (\kappa_{\text{obs}} - \bar{\kappa}_{\text{sim}})^2, \quad (3)$$

where $\text{ACF}_{\text{obs}}(\tau)$ and κ_{obs} are the empirical absolute-growth autocorrelation at lag τ and the global kurtosis, respectively (kurtosis denoted κ to avoid collision with the jump-duration variable K), and

Algorithm 1 HMM-WJ single asset marginal generator.

Require: Fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$; horizon T ; tail sets $\mathcal{S}_- = \{1, \dots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \dots, N\}$.
Ensure: State sequence $\{S_t\}_{t=0}^T$ and growth-rate path $\{G_t\}_{t=1}^T$.
1: Draw $S_0 \sim \bar{\pi}$; $k \leftarrow 0$ { k : remaining forced-step counter}
2: **for** $t = 1, \dots, T$ **do**
3: **if** $k > 0$ **then**
4: $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg} , else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow k - 1$ {forced tail step; sign drawn per step}
5: **else if** $\text{Uniform}(0, 1) > \epsilon$ **then** {standard Markov transition}
6: $S_t \sim \mathbf{T}_{S_{t-1}, \cdot}$
7: **else**
8: $K \sim \text{Poisson}(\lambda)$ {trigger jump; K is the episode length}
9: **if** $K = 0$ **then**
10: $S_t \sim \mathbf{T}_{S_{t-1}, \cdot}$ { $K = 0$: no forced steps, standard transition}
11: **else**
12: $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg} , else $\mathcal{S}_+$; $S_t \sim \text{Uniform}(\mathcal{J})$; $k \leftarrow \min(K - 1, T - t)$ {first forced step; K steps total, sign per step}
13: **end if**
14: **end if**
15: Draw $G_t \sim \mu_{S_t} + \sigma_{S_t} t_5$ {Eq. (2)}
16: **end for**

the simulated counterparts $\overline{\text{ACF}}_{\text{sim}}(\tau)$ and $\bar{\kappa}_{\text{sim}}$ were averages over jump-active paths. We simulated 200 independent paths of 2,766 trading days at each grid point. A path with no jump reduces to HMM-NJ and contains no information about (ϵ, λ) , so the objective included only paths with at least one jump [49]. The weight w_κ balanced the kurtosis and ACF terms. The search ranges, w_κ , L , and the remaining HMM-WJ and SIM hyperparameters are reported in Table 1. The fitted model was evaluated against the marginal and temporal statistics reported in Section 4.

3.2 Variance-corrected multi-asset composition

The single-asset model describes each ticker separately. It does not say how several tickers should move together. For the multi-asset extension, we added one common source of movement: a market path. We used the single-index model (SIM) of Sharpe [41]:

$$g_i(t) = \alpha_i + \beta_i g_m(t) + \varepsilon_i(t), \quad t = 1, \dots, T, \quad (4)$$

Here g_m is the market growth-rate path, β_i measures how strongly asset i moves with the market, and ε_i is the part specific to that asset. The symbol ε_i denotes a residual path; it is unrelated to the jump probability ϵ used by the single-asset model. We estimated (α_i, β_i) from real data by ordinary least squares (OLS). The remaining modeling choice is ε_i . A standard SIM uses Gaussian residuals with the variance implied by the real-data regression. This preserves the fitted market relationship, but it removes much of the tail and regime behavior produced by the single-asset generator. We called this baseline *Gaussian SIM*.

Why direct construction fails. A tempting alternative is to use a full synthetic return path $\tilde{g}_i$ from the single-asset generator as the residual:

$$g_i = \alpha_i + \beta_i g_m + \tilde{g}_i \quad (5)$$

where $\sigma_{\text{gen},i}^2 = \text{Var}(\tilde{g}_i)$. We called this the *naive* construction. Its problem is double counting. The draw $\tilde{g}_i$ already has the total variance of asset i . Adding the market term contributes more variance. Because $\tilde{g}_i$ is sampled independently of g_m ,

$$\text{Var}(g_i) = \beta_i^2 \sigma_m^2 + \sigma_{\text{gen},i}^2, \quad \text{Cov}(\tilde{g}_i, g_m) \approx 0. \quad (6)$$

The excess is $\beta_i^2 \sigma_m^2$, so the error is largest for assets with high market exposure.

Variance-preserving correction. We treated $\sigma_{\text{gen},i}^2$ as a variance budget for the composed asset. The market term uses part of that budget. We scaled the single-asset draw so that it uses only the part that remains. Set $\varepsilon_i = s_i \tilde{g}_i$ and require $\text{Var}(\beta_i g_m + s_i \tilde{g}_i) = \sigma_{\text{gen},i}^2$. Solving for s_i gives

$$s_i^2 = 1 - \frac{\beta_i^2 \sigma_m^2}{\sigma_{\text{gen},i}^2} \quad (7)$$

For clarity, define the *market loading ratio*

$$\rho_i \equiv \frac{\beta_i^2 \sigma_m^2}{\sigma_{\text{gen},i}^2}, \quad (8)$$

which is the fraction of the variance budget used by the market. The residual receives the remaining fraction, so $s_i^2 = 1 - \rho_i$. This correction is impossible when $\rho_i \geq 1$ because the market term alone uses the entire variance budget. This case can occur when β_i is large, the per-asset generator has low variance, or the synthetic market is more volatile than the calibration market. In that case we reserved a minimum fraction f for the asset-specific draw and reduced the effective beta. The rule is

$$\beta_i^{\text{eff}} = \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\ \text{sign}(\beta_i) \sqrt{(1-f) \frac{\sigma_{\text{gen},i}^2}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \quad s_i^2 = \begin{cases} 1 - \rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \quad (9)$$

The sign of β_i is retained. We used $f = 0.10$, which guarantees that at least 10% of the variance budget remains asset-specific (Table 1).

High- R^2 index trackers. The variance budget is the main target for ordinary stocks. For an index-tracking ETF, the fitted relationship with the market is more important. For example, SPY regressed on itself has $\beta = 1$, $R^2 = 1$, and no residual variance. QQQ and SPYG also have high R^2 against SPY. Applying the ordinary-stock rule to these trackers can preserve total variance while changing the fitted R^2 . We therefore used a separate branch when the real-data value $R_{i,\text{real}}^2$ is at least R_{preserve}^2 . In our universe, 0.80 separated the index trackers from individual stocks (Fig. S3). On this branch, the residual variance required to recover the calibrated R^2 is

$$\sigma_{\varepsilon,\text{target}}^2 = \beta_i^2 \sigma_m^2 \cdot \frac{1 - R_{i,\text{real}}^2}{R_{i,\text{real}}^2} \quad (10)$$

We rescaled the generator draw to this target:

$$\varepsilon_i = \left(\sqrt{\frac{\sigma_{\varepsilon,\text{target}}^2}{\sigma_{\text{gen},i}^2}} \right) \tilde{g}_i, \quad (11)$$

When $R_{i,\text{real}}^2 = 1$, the target residual variance is zero, as it should be for SPY against itself. If the scale factor in Eq. (11) exceeds one, the procedure stretches the generator draw. We flagged this case because it can weaken the original marginal fit.

Complete composition rule. Algorithm 2 puts these steps together. For a high- R^2 tracker, it preserves the calibrated R^2 . For other assets, it preserves the generator variance and clips beta only when the market term would leave too little asset-specific variance. It then adds the scaled asset draw to the market term. An optional rank reorder can add dependence among the residual draws without changing their individual variances. For long paths, OLS on the composed series recovers α_i and the effective beta. The composed variance matches the target selected by the branch. These are variance and factor-loading guarantees; they do not guarantee that the complete marginal distribution is unchanged. The algebra is given in Online Appendix S8. The construction adds only same-day market dependence. It does not reproduce lead-lag effects or market-driven volatility clustering.

Algorithm 2 Hybrid SIM composition with variance correction.

Require: Market growth-rate path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2 ; per-asset generator growth-rate paths $\tilde{g}_i \in \mathbb{R}^T$ for $i = 1, \dots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R_{i,\text{real}}^2)$; floor $f \in (0, 1)$; threshold $R_{\text{preserve}}^2 \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

```

1: for each asset  $i = 1, \dots, N$  do
2:   Compute generator variance  $\sigma_{\text{gen},i}^2 \leftarrow \text{Var}(\tilde{g}_i)$ .
3:   if  $R_{i,\text{real}}^2 \geq R_{\text{preserve}}^2$  then
4:      $\sigma_{\varepsilon,\text{target}}^2 \leftarrow \beta_i^2 \sigma_m^2 (1 - R_{i,\text{real}}^2) / R_{i,\text{real}}^2$  {Eq. (10)}
5:      $\varepsilon_i \leftarrow \left( \sqrt{\sigma_{\varepsilon,\text{target}}^2 / \sigma_{\text{gen},i}^2} \right) \tilde{g}_i$ 
6:      $\beta_i^{\text{eff}} \leftarrow \beta_i$ ; branch  $\leftarrow R^2$ -preserving
7:   else
8:      $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$  {Eq. (8)}
9:     if  $\rho_i \leq 1 - f$  then
10:       $s_i^2 \leftarrow 1 - \rho_i$ ;  $\beta_i^{\text{eff}} \leftarrow \beta_i$ ; branch  $\leftarrow$  variance-preserving
11:    else
12:       $s_i^2 \leftarrow f$ ;  $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i) \sqrt{(1 - f) \sigma_{\text{gen},i}^2 / \sigma_m^2}$ ; branch  $\leftarrow$  clipped variance-preserving
13:    end if
14:     $\varepsilon_i \leftarrow s_i \tilde{g}_i$ 
15:  end if
16:  (Optional) apply cross-sectional rank reorder to  $\varepsilon_i$ .
17:   $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}} g_m + \varepsilon_i$  {Eq. (4)}
18: end for

```

Evaluation protocol. We evaluated the method on 424 US equities and exchange-traded funds from Polygon.io. Each ticker had a complete daily-close history from January 2014 through December 2024 ($T = 2,767$). We used SPY as the market and treated the other 423 tickers as assets to compose. For each asset, we fitted the same state and emission model used in the SPY analysis (Table 1; $r_f = 0$). To focus the experiment on multi-asset composition rather than per-ticker jump calibration, we set the jump probability to zero. We also regressed each real asset series on SPY to estimate α_i , β_i , $R_{i,\text{real}}^2$, and residual variance. Across the universe, β ranged from 0.01 to 1.79 and $R_{i,\text{real}}^2$ ranged from 0.001 to 0.933 (Fig. S3).

We compared six ways to construct the asset-specific term. The naive method used the full JumpHMM draw without correction. The Gaussian SIM method used independent Gaussian residuals with variance estimated by OLS. The hybrid method used Algorithm 2. The JumpHMM-on-residuals method fitted a new JumpHMM to the OLS residual series after converting the residuals to a synthetic price path by cumulative compounding. The block-bootstrap method sampled the OLS residual series with the Politis–Romano stationary bootstrap [52] and a mean block length of $\sqrt{T} \approx 50$. The GARCH(1,1)- t method fitted a conditional-variance model to each residual series [7]. We excluded thirty non-stationary fits from the GARCH results only. For each ticker, the naive and hybrid methods received the same JumpHMM draw. This pairing isolated the effect of the variance correction. The other four methods generated their own residual draws. We did not use the optional cross-sectional rank reorder because the experiment tested each ticker’s marginal construction separately. We evaluated whether OLS recovered the calibrated β and R^2 , whether the synthetic series matched the real return distribution, and whether the composition preserved the generator variance. We measured these outcomes with $|\hat{\beta} - \beta|$, $|\hat{R}^2 - R_{\text{real}}^2|$, KS and AD tests, Wasserstein-1 distance, excess kurtosis, the Hill upper-tail index, and the ratio of composed to generator variance. Definitions are in Online Appendix S3. We ran 100 replications per ticker with seed 1234, producing 42,300 series per method. Pass rates used $\alpha = 0.05$.

We then froze all marginal fits, residual models, and SIM parameters estimated from 2014–2024 and evaluated the methods against the 249 growth rates observed in 2025. Of the 423 non-market training assets, 416 had complete holdout histories; GARCH results covered 386 because its thirty non-stationary training fits remained excluded. For each replication, we generated a 249-day SPY

path from the training-period SPY marginal and used that common simulated market path in all asset compositions. Observed 2025 SPY and asset returns were used only for scoring. We repeated the KS, AD, Wasserstein-1, variance, kurtosis, and one-day VaR evaluations with 100 paths per ticker. Because goodness-of-fit tests have less power on 249 observations than on 2,766, we also compared each synthetic path with a randomly selected contiguous 249-day block from the training history. We first averaged replication-level results within ticker and then summarized across tickers so each asset received equal weight.

4 Results

4.1 Marginal-generator validation

We first measured the empirical SPY data targets that any synthetic generator must reproduce. In both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 249$) windows, the real SPY series had excess kurtosis with Jarque-Bera rejection, persistent volatility clustering, and near-zero linear autocorrelation in raw returns (Table S1) [27]. We calibrated (ϵ, λ) for HMM-WJ against the in-sample SPY series using the grid search in Eq. (3) (Algorithm 1; search range in Table 1). The selected values were $(\epsilon, \lambda) = (10^{-4}, 100)$, although nearby grid points gave similar objective values. The primary analysis used $N = 100$ states. Across $N = 60$ to 200, the KS and AD pass rates varied by less than one percentage point for both HMM-NJ and HMM-WJ. At $N = 350$, some states were never visited and their emission parameters could not be estimated (Online Appendix S6).

With the model calibrated, we benchmarked HMM-WJ (Algorithm 1) against seven alternative generators on 1,000 synthetic paths of 2,766 trading days each (Table 2, Figure 3). Bootstrap resampling passed every marginal test but shared the i.i.d. autocorrelation error, while Gaussian sampling passed none. GARCH(1,1) had the lowest ACF-MAE (0.031) but rarely passed the marginal tests, and the Gated Recurrent Unit (GRU) [53] behaved similarly, with competitive temporal error and near-zero marginal pass rates. The hidden semi-Markov model (HSMM) [16] reached 82% KS but underestimated kurtosis by more than a third. A comparison of persistence specifications (HMM-WJ vs HSMM) and emission families (Gaussian vs Student- t) is given in Table S3 (Online Appendix S5). HMM-NJ and HMM-WJ had KS pass rates of 99.7% and 97.6%, respectively; HMM-WJ’s pass rates ran 2–8 percentage points below HMM-NJ’s because jumps occasionally overweight tail states relative to their empirical frequency. The Student- t ($\nu = 5$) emission produced simulated kurtosis of 7.6 against the observed 7.7 (Fig. 3c), while Normal emissions underestimated it by $\sim 29\%$ (Table S3). Both HMM variants reproduced the observed negative skewness of -0.75 (Table S1), which came from asymmetric occupancy of the Laplace states rather than an explicit skewness parameter; the symmetric-innovation models (GARCH, Gaussian, and Laplace) produced near-zero skewness by construction. For temporal dependence, HMM-WJ reduced ACF-MAE from 0.059 for HMM-NJ to 0.052, and only HMM-WJ, among the non-GARCH, non-neural models, scored below the value shared by the i.i.d. samplers, driven by the $\sim 24\%$ of paths that contained at least one Poisson jump and sustained ACF($|G_t|$) decay across all lags; the jump override was inactive in the remaining paths, for which HMM-WJ reduces to HMM-NJ. On the held-out 249 days of 2025, HMM-NJ and HMM-WJ achieved 96.7% and 94.4% KS with simulated kurtosis close to the observed value, while GARCH’s W_1 rose by 67% from its in-sample value (Table 2; Figure S4). Cross-asset evaluation on NVDA, JNJ, and JPM produced HMM-WJ in-sample KS pass rates of at least 98.9% on all three, with out-of-sample rates of 97.7%, 74.0%, and 79.2%, respectively (Table S7, Online Appendix S7).

To measure how jump-active paths changed the results, we resampled the fitted ensemble at different jump-path fractions f (Figure 4). Here f is the share of accepted paths that contain at least one jump. We varied f from 0, where all accepted paths reduce to HMM-NJ, to 1, where every accepted path contains a jump. This procedure held the fitted model fixed and changed only the mixture of accepted paths. The absolute-return autocorrelation error fell monotonically from 0.059 at $f = 0$ to 0.037 at $f = 1$, moving from the no-jump value toward the GARCH value as the jump share rose. Over the same sweep, simulated excess kurtosis declined from 7.9 to 6.7, matching the observed value near $f \approx 0.14$ and falling below it thereafter. The two-sample KS and AD pass rates also declined. The Hill upper-tail index stayed near 0.35, against an observed 0.32, across the whole sweep because the Student- t ($\nu = 5$) emission sets the tail exponent independently of jump content. Increasing the jump share therefore traded fit to the distributional body for lower autocorrelation error without changing the tail index. The value $f \approx 0.25$ used throughout kept simulated kurtosis within 2% of

the observed value and lowered the autocorrelation error about a quarter of the way from the no-jump value toward the all-jump endpoint.

4.2 Variance-corrected multi-asset composition

The single-asset results defined the tradeoff in the marginal generator. We then tested whether the variance-corrected SIM preserved each asset’s variance and factor loading across the 423-ticker universe, comparing the six composition methods on the aggregate scorecard (Table 3). The median absolute β error ranged from 0.018 to 0.022 across methods (Table 3); the methods differed more in R^2 recovery, marginal fit, and tail behavior. Gaussian SIM had the lowest median R^2 error because it targets R^2 by construction, but it almost never passed the marginal tests and produced the lightest tails of the six methods, with excess kurtosis an order of magnitude below the real data (Fig. S2). Naive composition retained the generator marginal but inflated its variance by $\beta_i^2 \sigma_m^2$, so it too failed nearly all marginal tests. The hybrid method passed 50.4% of KS replications and preserved the generator’s tail behavior, with median excess kurtosis of 7.2 and a Hill index of 0.36; its median R^2 error sat between the naive and Gaussian extremes because the variance-preserving branch targets $\sigma_{\text{gen},i}^2$ rather than $R_{i,\text{real}}^2$. The three residual-fit methods sidestepped the variance-inflation problem entirely by fitting a generator directly to the empirical OLS residual series, whose variance excludes the market component by construction. Their KS pass rates ran from 66.9% (GARCH(1,1)- t) to 75.8% (JumpHMM-on-residuals), higher than the hybrid’s, with correspondingly lower Wasserstein distances. JumpHMM-on-residuals matched the hybrid’s kurtosis and Hill index almost exactly, while block bootstrap and GARCH, by fitting their own residual generators, produced lighter tails than either JumpHMM-based method.

The same ordering largely persisted when the fitted models and SIM parameters were frozen and evaluated against 416 complete asset histories from 2025 (Table 4). The hybrid method reached an 84.1% mean cross-ticker KS pass rate, against 78.8% for naive composition, and its median composed-to-observed variance ratio was 0.97, whereas naive composition remained inflated at 1.29. JumpHMM-on-residuals retained the strongest marginal fit, with an 86.2% KS pass rate and a slightly lower median W_1 than the hybrid. Matched-length training blocks produced lower non-rejection rates for every method, showing that the higher 2025 pass rates did not arise solely from comparing samples of 249 rather than 2,766 observations. These averages nevertheless concealed cross-ticker heterogeneity (Figure 5): most ticker-level pass rates concentrated near the upper end of the range, but every method retained a lower tail of weak fits, and for the hybrid method many tickers improved relative to their matched-length training blocks while a subset deteriorated out of sample.

4.3 VaR coverage

We used the frozen synthetic paths to estimate per-ticker one-day Value-at-Risk (VaR) thresholds and counted exceedances on the real 2025 histories (Table 5). At $\alpha = 0.95$, the hybrid and JumpHMM-on-residuals methods produced mean exceedance rates of 6.13% and 6.07%, respectively, against the 5% nominal rate. At $\alpha = 0.99$, their rates were 1.71% and 1.67%, against the 1% nominal rate. Naive composition was closest at the 99% level (1.24%), while Gaussian SIM had the largest exceedance rate (2.03%). The short holdout contains only about 2–3 expected 99% exceedances per ticker, so we interpret the corresponding Kupiec pass rates and cross-ticker dispersion as low-power checks rather than precise rankings.

4.4 Sensitivity analyses

To isolate how the variance correction changed marginal fit, we plotted each ticker’s KS pass rate and variance ratio $\text{Var}(g)/\sigma_{\text{gen}}^2$ against its calibrated β (Fig. 6). Under naive composition, the variance ratio followed $1 + \rho_i$ and exceeded generator variance by 50–75% at high β , and the naive KS pass rate fell toward zero above $\beta \approx 0.7$; the hybrid method held the ratio at one and kept per-ticker pass rates between 0.3 and 0.9 across the β range. The branch-selection rule assigned 421 tickers to the variance-preserving branch, two trackers to the R^2 -preserving branch, and none to the clipped branch (Fig. S6, Table S4). The two R^2 -preserving assignments were QQQ and SPYG, and both had a 99% KS pass rate; the highest- R^2 individual stock, BLK, fell about 0.16 below the $R_{\text{preserve}}^2 = 0.80$ threshold. Clipping did not occur in the universe: $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ (Eq. (8)) stayed below $1 - f = 0.90$ for every ticker. Because the R^2 -preserving branch contained only two tickers, we tested it on

a synthetic-tracker grid spanning $(\beta_{\text{true}}, R_{\text{true}}^2) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$. Every cell passed KS.

Bucketing the universe by β quartile, the hybrid KS pass rate fell from 67% in the lowest quartile to 44% in the highest (Table S5). As β increased, the market term accounted for more of the composed variance, making the marginal more dependent on the market distribution. We tested the clipping rule directly by rescaling the market growth-rate path by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed (Table S6). At $\gamma = 1$ no ticker crossed the threshold, and the hybrid KS pass rate was 50.3%. At $\gamma = 2$ and 3, clipping applied to 76.4% and 95.2% of tickers, and the KS pass rate rose to 71.8% and 77.3%.

Finally, we tested the aggregate hybrid KS pass rate of Table 3 against the two hyperparameter choices of Algorithm 2 (Table 1), re-running hybrid on the full 423-ticker universe at every point of the $R_{\text{preserve}}^2 \times f$ grid in Fig. S7. The hybrid KS pass rate varied by 0.14 percentage points across the 25 grid cells. The two index trackers remained in the R^2 -preserving branch for every threshold at or below 0.85, and the floor f did not change any branch assignment. We then measured the contribution of generator stochasticity to the aggregate metrics by re-running the full six-method protocol under 8 random seeds (Table S8). Per-method KS pass rates held to within fractions of a percentage point across seeds, and the seed sweep reproduced the relative ordering of the methods without inversions. For example, the 25 percentage-point KS difference between hybrid and JumpHMM-on-residuals exceeded hybrid’s across-seed SD of 0.18 percentage points.

5 Discussion

The single-asset analysis supplied the marginal input for the multi-asset construction. Relative to HMM-NJ, HMM-WJ reduced ACF-MAE at the cost of somewhat lower in-sample KS and AD pass rates. GARCH reached the lowest ACF-MAE of any generator but failed the distributional tests, and the GRU behaved the same way. The HSMM passed the KS test but underproduced kurtosis relative to the observed data. Bootstrap resampling passed KS trivially by drawing only observed returns. Among the models that were neither GARCH nor neural, HMM-WJ was the only one with ACF-MAE below the i.i.d. baseline, and the same ordering held in the 2025 evaluation. We therefore treated HMM-WJ as a marginal generator that traded lower distributional pass rates for lower ACF-MAE, not as a uniformly better single-asset model. These pass-rate comparisons have a statistical limitation. Two-sample KS and AD p -values assume independent observations, but the HMM, HSMM, GARCH, and bootstrap generators produce dependent observations. We therefore considered the pass rates alongside W_1 , Hellinger distance, and the Hill tail index rather than comparing their p -values on the same footing. The $N = 100$ state resolution, selected by a sensitivity sweep over $\{30, 60, 90, 100, 150, 200\}$, produced little KS or AD variation across the middle of that range on the 2,766-day SPY sample. Shorter histories will support fewer states because each state will receive fewer observations. The jump parameters (ϵ, λ) were tuned once against SPY and applied across the universe. Per-ticker tuning on the three cross-asset tickers produced in-sample KS pass rates of at least 98.9% (Online Appendix S7). These three cases do not show that the SPY calibration will transfer to assets with different jump behavior. The Student- t emission with $\nu = 5$ matched the observed kurtosis far more closely than Normal emissions, but it does not model skewness directly. The negative skewness in both HMM variants came from asymmetric occupancy of the Laplace states. Assets with a different skew direction may require a different quantile rule.

The main result was the multi-asset variance correction. Naive composition inflated generator variance at high β and almost always failed the KS test, as did the Gaussian SIM. The corrected method held the variance ratio at one and passed the KS test about half the time in the full-length training comparison. With all fits frozen for 2025, it retained a median variance ratio of 0.97 and improved the mean cross-ticker KS pass rate from 78.8% under naive composition to 84.1%. This improvement was not uniform across assets: Figure 5 shows generally strong ticker-level pass rates alongside a lower tail of weak fits, with some hybrid tickers deteriorating relative to matched-length training blocks. The correction therefore improved typical marginal fit and variance calibration without eliminating asset-specific distribution shift. This heterogeneity makes ticker-level diagnostics important; a universe-wide average pass rate should not be interpreted as a deployment guarantee for every asset. The residual-fit methods had higher KS pass rates than the hybrid method in both windows. These methods fit the OLS residuals directly and therefore avoid the variance-inflation problem. The choice between the two approaches depends on whether the same full-return generator

must serve both single-asset and multi-asset uses. The hybrid method’s held-out one-day 99% VaR exceedance rate was 1.71%, comparable to 1.67% for JumpHMM-on-residuals but above the 1% nominal level. It did not require the residual method’s separate fitting stage, though its marginal KS pass rate was lower.

Several evaluation choices limit these results. The in-sample composition experiment fixed the market path at the real SPY history; the holdout experiment instead generated SPY from its frozen training-period marginal and reserved observed 2025 SPY for scoring. The naive and hybrid methods used the same JumpHMM draw for each ticker, while the other methods drew from their own generators. We also omitted the optional cross-sectional rank reorder and evaluated VaR per ticker rather than at the portfolio level. Portfolio-level VaR would require a model of cross-asset dependence. The variance correction assumes $\text{Cov}(\tilde{g}_i, g_m) \approx 0$, which holds by construction here because the JumpHMM marginals were fitted and sampled independently of the market path. The correction would need adjustment for a generator already correlated with the market (for example, a bootstrap from joint history). The construction also injects contemporaneous market dependence only: lagged effects such as lead-lag correlation or market-driven volatility clustering are not reproduced. Clipping did not trigger because every ticker had $\rho_i < 0.90$. It becomes relevant when the simulated market is more volatile than the calibration history or for universes mixing low-volatility stocks with extreme-beta assets such as leveraged ETFs. Even without clipping, the hybrid KS pass rate declined from the lowest β quartile to the highest. As β increased, the market term accounted for more of the composed distribution, whose tails then depended more on the market path. The 249-day holdout also has substantially less power than the 2,766-day training comparison. Matched-length training blocks partly address this difference, but neither non-rejection rates nor 99% VaR counts from a single year should be interpreted as precise model rankings.

Independent residuals produced a diversification-return bias under daily rebalancing. Matching each asset’s marginal variance did not restore the residual covariance created by common sector and style factors [45]. The resulting excess cross-sectional dispersion became a spurious return premium under rebalancing. This problem affected every method that drew residuals independently, not only naive composition. The block-bootstrap and GARCH methods restored each ticker’s temporal structure but not covariance between tickers. In a 20-asset target-weight allocator example, the synthetic median annualized growth rate exceeded the real 2025 return by about 18 percentage points. The static-versus-daily-rebalance difference was 27.73 percentage points, whereas adding the allocation signal changed the median by 0.13 percentage points [45, 54, 55] (Online Appendix S2). We applied a location-shift correction that subtracts a constant drift in log-wealth from every path, calibrated to match an external long-run prior [56, 57] rather than the realized test-year outcome. The correction preserved path ranks but changed drawdown dates and depths because the adjustment varies over time in wealth space. It recalibrated the return level but did not restore cross-asset covariance. The full method, the rejected alternatives, and the percentile placement of the realized year are given in Online Appendix S2.

Several extensions remain open. Per-ticker tuning of the jump parameters (ϵ, λ) on each asset’s own history would require a separate calibration for every asset. Multi-factor or lagged-factor variants could add the dependence omitted by the current single-factor construction. The same variance correction extends to either case, but we did not evaluate those extensions. A joint residual model, such as a synchronized multivariate block bootstrap, a residual factor model, or a copula, could restore cross-asset residual covariance. In contrast, separate autoregressive or block-bootstrap models would improve each asset’s temporal dependence without creating dependence between assets. A stress test that pairs the clipping rule with a market more volatile than the calibration history would test whether the idiosyncratic floor keeps per-asset marginals heavy-tailed under exactly the conditions where naive SIM composition breaks down.

6 Conclusion

In this study, we developed a variance-corrected method for composing per-asset generators through a single-index model. The correction removed the variance inflation caused by inserting a full-return generator draw into the market factor. Across 423 non-market assets, it recovered the calibrated factor loading to a low median absolute error and produced heavy-tailed marginals. On 416 complete held-out asset histories from 2025, it improved the mean cross-ticker KS pass rate from 78.8% under naive composition to 84.1% and held the median variance ratio at 0.97. Its one-day 99%

Value-at-Risk exceedance rate was 1.71%, compared with 1.67% for JumpHMM-on-residuals and a 1% nominal rate. Those residual models, fitted directly to the OLS residual series, reached higher KS pass rates than the corrected method but required a separate fit for each asset. The per-asset component underpinning this construction was a hidden Markov marginal model with Laplace-quantile states, Student- t emissions, and direct transition counting. On SPY, its jump-duration variant reduced absolute-return autocorrelation error relative to the no-jump variant while keeping high KS and AD pass rates; GARCH reached a lower autocorrelation error but failed the distributional tests. The single-asset result was therefore a measured marginal–temporal tradeoff rather than a claim of uniform superiority.

The current multi-asset implementation does not model cross-asset residual covariance. In a 20-asset allocator, independent residuals inflated the synthetic median daily-rebalanced return well above the realized 2025 outcome, almost entirely through the static-to-daily-rebalance diversification return rather than the allocation signal. A location-shift correction set to a documented long-run prior adjusted the return level without restoring the missing covariance. A full joint residual model is needed before using these paths for portfolio-level return forecasts.

Acknowledgments

The authors thank the eCornell AI in Finance program participants for discussions that motivated the downstream-use bias-correction analysis, and Polygon.io for access to the 2014 to 2024 US equity and ETF daily price histories used in the experiments. Computations were run on a single laptop; no shared-resource allocation was required.

Author contributions. A.A. designed the hybrid hidden Markov marginal generator, performed the single-asset experiments and validation framework, and contributed to the manuscript. J.D.V. designed the variance-corrected single-index composition, performed the multi-asset experiments and downstream Value-at-Risk validation, developed the bias-correction methodology, released the code, and contributed to the manuscript.

Competing interests. The authors declare no competing interests.

Code and data availability. The experiment scripts, evaluation harness (six-composer benchmark, Value-at-Risk coverage check, bias-correction utility), and the fitted per-asset marginals are released in the paper repository (<https://github.com/varnerlab/HMM-w-jumps-paper>); the underlying model-fitting library is released separately as JumpHMM.jl (<https://github.com/varnerlab/JumpHMM.jl>), pinned as a dependency of the experiment code.

References

- [1] Samuel A. Assefa, Danial Dervovic, Mahmoud Mahfouz, Robert E. Tillman, Prashant Reddy, and Manuela Veloso. Generating synthetic data in finance: Opportunities, challenges and pitfalls. In *Proceedings of the First ACM International Conference on AI in Finance*, pages 1–8, 2020. doi: 10.1145/3383455.3422554.
- [2] James Jordon, Lukasz Szpruch, Florimond Houssiau, Mirko Bottarelli, Giovanni Cherubin, Carsten Maple, Samuel N. Cohen, and Adrian Weller. Synthetic data: What, why and how? *arXiv preprint arXiv:2205.03257*, 2022.
- [3] Benoit Mandelbrot. The variation of certain speculative prices. *The Journal of Business*, 36(4): 394–419, 1963.
- [4] R. Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001. doi: 10.1080/713665670.
- [5] Michael Stenger, André Bauer, Thomas Prantl, Robert Leppich, Nathaniel Hudson, Kyle Chard, Ian Foster, and Samuel Kounev. Thinking in categories: A survey on assessing the quality for time series synthesis. *Journal of Data and Information Quality*, 16(2):1–32, 2024. doi: 10.1145/3666006.

- [6] Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.
- [7] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.
- [8] Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993. doi: 10.1093/rfs/6.2.327.
- [9] Robert C. Merton. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1-2):125–144, 1976. doi: 10.1016/0304-405X(76)90022-2.
- [10] S. G. Kou. A jump-diffusion model for option pricing. *Management Science*, 48(8):1086–1101, 2002. doi: 10.1287/mnsc.48.8.1086.166.
- [11] Shuntaro Takahashi, Yu Chen, and Kumiko Tanaka-Ishii. Modeling financial time-series with generative adversarial networks. *Physica A: Statistical Mechanics and Its Applications*, 527: 121261, 2019. doi: 10.1016/j.physa.2019.121261.
- [12] Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [13] Sohyeon Kwon and Yongjae Lee. Can GANs learn the stylized facts of financial time series? In *Proceedings of the 5th ACM International Conference on AI in Finance*, pages 126–133, 2024. doi: 10.1145/3677052.3698661.
- [14] L. Rabiner and B. Juang. An introduction to hidden Markov models. *IEEE ASSP Magazine*, 3(1):4–16, 1986. doi: 10.1109/MASSP.1986.1165342.
- [15] Tobias Rydén, Timo Teräsvirta, and Stefan Åsbrink. Stylized facts of daily return series and the hidden Markov model. *Journal of Applied Econometrics*, 13(3):217–244, 1998. doi: 10.1002/(SICI)1099-1255(199805/06)13:3<217::AID-JAE476>3.0.CO;2-V.
- [16] Jan Bulla and Ingo Bulla. Stylized facts of financial time series and hidden semi-Markov models. *Computational Statistics & Data Analysis*, 51(4):2192–2209, 2006. doi: 10.1016/j.csda.2006.07.021.
- [17] Thomas Lux and Michele Marchesi. Scaling and criticality in a stochastic multi-agent model of a financial market. *Nature*, 397:498–500, 1999. doi: 10.1038/17290.
- [18] Rama Cont and Jean-Philippe Bouchaud. Herd behavior and aggregate fluctuations in financial markets. *Macroeconomic Dynamics*, 4(2):170–196, 2000. doi: 10.1017/S1365100500015029.
- [19] J. Doyne Farmer and Shareen Joshi. The price dynamics of common trading strategies. *Journal of Economic Behavior & Organization*, 49(2):149–171, 2002. doi: 10.1016/S0167-2681(02)00065-3.
- [20] Blake LeBaron. Agent-based computational finance. In Leigh Tesfatsion and Kenneth L. Judd, editors, *Handbook of Computational Economics*, volume 2, pages 1187–1233. Elsevier, 2006. doi: 10.1016/S1574-0021(05)02024-1.
- [21] W. Brian Arthur. Foundations of complexity economics. *Nature Reviews Physics*, 3:136–145, 2021. doi: 10.1038/s42254-020-00273-3.
- [22] J. Doyne Farmer. Quantitative agent-based models: A promising alternative for macroeconomics. *Oxford Review of Economic Policy*, 41(2):571–590, 2025. doi: 10.1093/oxrep/graf027.
- [23] Louis Bachelier. Théorie de la spéculation. *Annales Scientifiques de l’École Normale Supérieure*, 17:21–86, 1900.
- [24] M. F. M. Osborne. Brownian motion in the stock market. *Operations Research*, 7(2):145–173, 1959.

- [25] Paul A. Samuelson. Rational theory of warrant pricing. *Industrial Management Review*, 6(2): 13–32, 1965.
- [26] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.
- [27] Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970. doi: 10.2307/2325486.
- [28] G. William Schwert. Why does stock market volatility change over time? *The Journal of Finance*, 44(5):1115–1153, 1989. doi: 10.1111/j.1540-6261.1989.tb02647.x.
- [29] Elias M. Stein and Jeremy C. Stein. Stock price distributions with stochastic volatility: An analytic approach. *The Review of Financial Studies*, 4(4):727–752, 1991.
- [30] Sangjoon Kim, Neil Shephard, and Siddhartha Chib. Stochastic volatility: Likelihood inference and comparison with ARCH models. *The Review of Economic Studies*, 65(3):361–393, 1998.
- [31] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi: 10.2307/1912559.
- [32] Alessandro Rossi and Giampiero M. Gallo. Volatility estimation via hidden Markov models. *Journal of Empirical Finance*, 13(2):203–230, 2006. doi: 10.1016/j.jempfin.2005.09.003.
- [33] Nguyet Nguyen. Hidden Markov model for stock trading. *International Journal of Financial Studies*, 6(2):36, 2018. doi: 10.3390/ijfs6020036.
- [34] Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970. doi: 10.1214/aoms/1177697196.
- [35] José G. Dias, Jeroen K. Vermunt, and Sofia Ramos. Mixture hidden Markov models in finance research. In *Advances in Data Analysis, Data Handling and Business Intelligence*, pages 451–459. Springer, 2010. doi: 10.1007/978-3-642-01044-6_41.
- [36] Paul Embrechts, Alexander J. McNeil, and Daniel Straumann. Correlation and dependence in risk management: Properties and pitfalls. In *Risk Management: Value at Risk and Beyond*, pages 176–223. Cambridge University Press, 2002. doi: 10.1017/CBO9780511615337.008.
- [37] Umberto Cherubini, Elisa Luciano, and Walter Vecchiato. *Copula Methods in Finance*. John Wiley & Sons, 2004. ISBN 9780470863459.
- [38] Kjersti Aas, Claudia Czado, Arnoldo Frigessi, and Henrik Bakken. Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2):182–198, 2009.
- [39] Stephen A. Ross. The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3):341–360, 1976. doi: 10.1016/0022-0531(76)90046-6.
- [40] Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993. doi: 10.1016/0304-405X(93)90023-5.
- [41] William F. Sharpe. A simplified model for portfolio analysis. *Management Science*, 9(2): 277–293, 1963. doi: 10.1287/mnsc.9.2.277.
- [42] A. N. Kolmogorov. Sulla determinazione empirica di una legge di distribuzione. *Giornale dell’Istituto Italiano degli Attuari*, 4:83–91, 1933.
- [43] N. Smirnov. Table for estimating the goodness of fit of empirical distributions. *The Annals of Mathematical Statistics*, 19(2):279–281, 1948.
- [44] Theodore W. Anderson and Donald A. Darling. Asymptotic theory of certain “goodness of fit” criteria based on stochastic processes. *The Annals of Mathematical Statistics*, 23(2):193–212, 1952.

- [45] David G. Booth and Eugene F. Fama. Diversification returns and asset contributions. *Financial Analysts Journal*, 48(3):26–32, 1992. doi: 10.2469/faj.v48.n3.26.
- [46] Samuel Kotz, Tomasz J. Kozubowski, and Krzysztof Podgórski. *The Laplace Distribution and Generalizations: A Revisit with Applications to Communications, Economics, Engineering, and Finance*. Birkhauser, 2001. ISBN 9780817641665.
- [47] David Toth and Bruce Jones. Against the norm: Modeling daily stock returns with the Laplace distribution. *arXiv preprint arXiv:1906.10325*, 2019.
- [48] Jeff A. Bilmes. A gentle tutorial of the EM algorithm and its application to parameter estimation for Gaussian mixture and hidden Markov models. Technical Report TR-97-021, International Computer Science Institute, 1998.
- [49] Paul Glasserman. *Monte Carlo Methods in Financial Engineering*. Springer, 2003. doi: 10.1007/978-0-387-21617-1.
- [50] James D. Hamilton. Regime switching models. *The New Palgrave Dictionary of Economics*, pages 11421–11426, 2018. doi: 10.1057/978-1-349-95189-5_2459.
- [51] Charles M. Grinstead and J. Laurie Snell. *Introduction to Probability*. American Mathematical Society, 1997.
- [52] Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870.
- [53] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734. Association for Computational Linguistics, 2014. doi: 10.3115/v1/D14-1179.
- [54] Paul Bouchev, Vassilii Nemtchinov, and Ting-Kam Leonard Wong. Volatility harvesting in theory and practice. *Journal of Wealth Management*, 18(3):89–100, 2015. doi: 10.3905/jwm.2015.18.3.089.
- [55] Harry M. Markowitz. Investment for the long run: New evidence for an old rule. *Journal of Finance*, 31(5):1273–1286, 1976. doi: 10.1111/j.1540-6261.1976.tb03213.x.
- [56] Aswath Damodaran. Equity risk premiums: Determinants, estimation and implications. NYU Stern School of Business working paper, updated annually, 2025. <https://pages.stern.nyu.edu/~adamodar/>.
- [57] Roger G. Ibbotson and Duff and Phelps. *SBBI Yearbook: Stocks, Bonds, Bills, and Inflation*. Duff and Phelps / Kroll, annual edition, 2024.

Table 1: **HMM-WJ and SIM composition hyperparameters.** Values used for the main SPY analysis, with the sensitivity-sweep or grid-search span and the location of the corresponding check where one exists. The 423-ticker SIM composition universe holds $\epsilon = 0$ instead of the SPY value; per-ticker (ϵ^*, λ^*) for the NVDA, JNJ, and JPM cross-asset validation are given in Table S7.

Symbol	Description	SPY value	Explored range
<i>State partition and emission</i>			
N	Laplace-quantile states	100	$\{30, \dots, 350\}$, Online Appendix S6
ν	Student- t emission degrees of freedom	5	$\{3, \dots, 30, \infty\}$
<i>Jump-duration override</i>			
ϵ	jump probability per step	10^{-4}	$[10^{-4}, 2.5 \times 10^{-2}]$, Eq. (3)
λ	mean jump-episode duration (steps)	100	$[10, 160]$, Eq. (3)
p_{neg}	probability a jump targets $\mathcal{S}_-$	0.52	fixed
N_{tail}	quantile states per tail set	5	fixed (at $N = 100$)
w_κ	kurtosis-penalty weight, Eq. (3)	0.20	fixed
L	ACF lag window, Eq. (3)	25 days	fixed
<i>SIM composition</i>			
f	idiosyncratic variance floor	0.10	$\{0.05, \dots, 0.30\}$, Fig. S7
R_{preserve}^2	branch-selection threshold	0.80	$\{0.70, \dots, 0.90\}$, Fig. S7

Table 2: **In-sample and out-of-sample model comparison for SPY.** Results summarize 1,000 simulated paths. Values in parentheses are standard errors, and bold entries are best within each panel. Metric definitions and standard-error calculations are given in Online Appendix S3.

Model	KS (%)	AD (%)	κ	ACF-MAE	Cov. (%)	W_1	Hellinger
<i>In sample: 2014–2024 (2,766 days; observed $\kappa = 7.715$)</i>							
Bootstrap	100.0 (<0.1)	99.7 (0.2)	7.6 (0.08)	0.060 (<0.001)	100.0 (<0.1)	0.062 (0.001)	0.042 (<0.001)
Gaussian	0.0 (<0.1)	0.0 (<0.1)	−0.0 (<0.01)	0.060 (<0.001)	13.1 (0.1)	0.399 (0.001)	0.148 (<0.001)
Laplace	44.0 (1.6)	43.3 (1.6)	3.0 (0.02)	0.060 (<0.001)	37.4 (1.5)	0.138 (0.001)	0.072 (<0.001)
GARCH(1,1)	5.5 (0.7)	1.9 (0.4)	8.2 (0.37)	0.031 (0.001)	29.3 (1.4)	0.304 (0.006)	0.100 (0.001)
GRU	0.6 (0.2)	0.2 (0.1)	5.7 (0.16)	0.036 (<0.001)	17.2 (0.5)	0.421 (0.003)	0.134 (<0.001)
HSMM	82.0 (1.2)	42.5 (1.6)	4.8 (0.08)	0.059 (<0.001)	68.7 (1.1)	0.176 (0.001)	0.113 (<0.001)
HMM-NJ	99.7 (0.2)	99.1 (0.3)	8.1 (0.14)	0.059 (<0.001)	100.0 (<0.1)	0.081 (0.001)	0.073 (<0.001)
HMM-WJ	97.6 (0.5)	91.3 (0.9)	7.6 (0.14)	0.052 (<0.001)	100.0 (<0.1)	0.101 (0.001)	0.075 (<0.001)
<i>Out of sample: 2025 (249 days; observed $\kappa = 6.867$)</i>							
Bootstrap	97.4 (0.5)	98.5 (0.4)	6.0 (0.18)	0.043 (<0.001)	100.0 (<0.1)	0.232 (0.002)	0.205 (0.001)
Gaussian	62.2 (1.5)	46.3 (1.6)	−0.0 (<0.01)	0.043 (<0.001)	36.4 (1.9)	0.452 (0.002)	0.235 (0.001)
Laplace	88.0 (1.0)	92.9 (0.8)	2.7 (0.05)	0.043 (<0.001)	74.7 (1.0)	0.263 (0.002)	0.211 (0.001)
GARCH(1,1)	80.3 (1.3)	72.0 (1.4)	1.6 (0.07)	0.026 (<0.001)	96.0 (1.3)	0.507 (0.018)	0.232 (0.001)
GRU	71.8 (1.4)	44.8 (1.6)	2.9 (0.10)	0.033 (<0.001)	77.8 (2.9)	0.488 (0.005)	0.240 (0.001)
HSMM	96.2 (0.6)	96.7 (0.6)	4.1 (0.13)	0.042 (<0.001)	100.0 (0.2)	0.287 (0.002)	0.239 (0.001)
HMM-NJ	96.7 (0.6)	96.7 (0.6)	6.4 (0.22)	0.041 (<0.001)	100.0 (<0.1)	0.258 (0.002)	0.207 (0.001)
HMM-WJ	94.4 (0.7)	95.1 (0.7)	6.1 (0.13)	0.039 (<0.001)	100.0 (<0.1)	0.282 (0.006)	0.210 (0.001)

KS and AD report pass rates at $\alpha = 0.05$; coverage is the fraction of empirical quantiles inside the synthetic 90% envelope. κ is simulated excess kurtosis. ACF-MAE is computed for $|G_t|$ through lag 252. Larger values are preferred for KS, AD, and coverage; smaller values are preferred for ACF-MAE, W_1 , and Hellinger distance; κ is compared with the observed value.

Table 3: **Aggregate scorecard across 423 non-market tickers and 100 replications per ticker.** Median values per composition method. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R_{\text{real}}^2|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1 : Wasserstein-1 distance to the real marginal. κ : sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$.

Composer	$ \hat{\beta} - \beta $	$ \hat{R}^2 - R_{\text{real}}^2 $	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365
JumpHMM-on-residuals	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)- t	0.017	0.036	67.4	61.4	0.267	6.069	0.317

Table 4: **Frozen-parameter multi-asset evaluation on the 2025 holdout.** Results cover 416 non-market tickers and 100 replications per ticker; GARCH covers 386 tickers. Pass rates are first averaged within ticker and then across tickers. *matched IS* compares each synthetic path with a randomly selected contiguous 249-day block from the 2014–2024 training history, matching the holdout sample length. W_1 and the variance ratio are medians across ticker-level medians; the variance ratio is synthetic variance divided by observed 2025 variance.

Composer	KS pass (%)		AD pass (%)		W_1 (2025)	variance ratio
	matched IS	2025	matched IS	2025		
Naive	65.7	78.8	51.6	67.1	0.914	1.29
Gaussian SIM	57.4	73.9	44.9	64.6	0.866	0.97
Hybrid	76.5	84.1	64.0	76.4	0.741	0.97
JumpHMM-on-residuals	77.9	86.2	65.6	80.1	0.708	0.98
Block bootstrap	76.3	84.0	64.0	76.5	0.715	0.90
GARCH(1,1)- t	74.5	82.6	62.6	75.0	0.733	0.91

Table 5: **Out-of-sample per-ticker one-day Value-at-Risk coverage across the six composition methods.** VaR thresholds at coverage level α were estimated from frozen-parameter synthetic paths and exceedances were counted against the real 2025 growth-rate history. *rate*: mean of the per-ticker mean exceedance rates (nominal: $1 - \alpha$). *SD*: cross-ticker standard deviation, in percentage points. *Kupiec pass*: mean within-ticker fraction of replications whose unconditional-coverage test p -value exceeds 0.05.

Composer	$\alpha = 0.95$			$\alpha = 0.99$		
	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	4.51	2.17	65.7	1.24	0.77	78.3
Gaussian SIM	4.95	2.26	70.0	2.03	1.17	72.1
Hybrid	6.13	2.45	64.2	1.71	0.94	72.5
JumpHMM-on-residuals	6.07	2.59	67.0	1.67	0.95	74.5
Block bootstrap	6.13	2.56	63.7	1.84	1.01	73.4
GARCH(1,1)- t	5.79	2.47	62.4	1.80	1.02	72.0

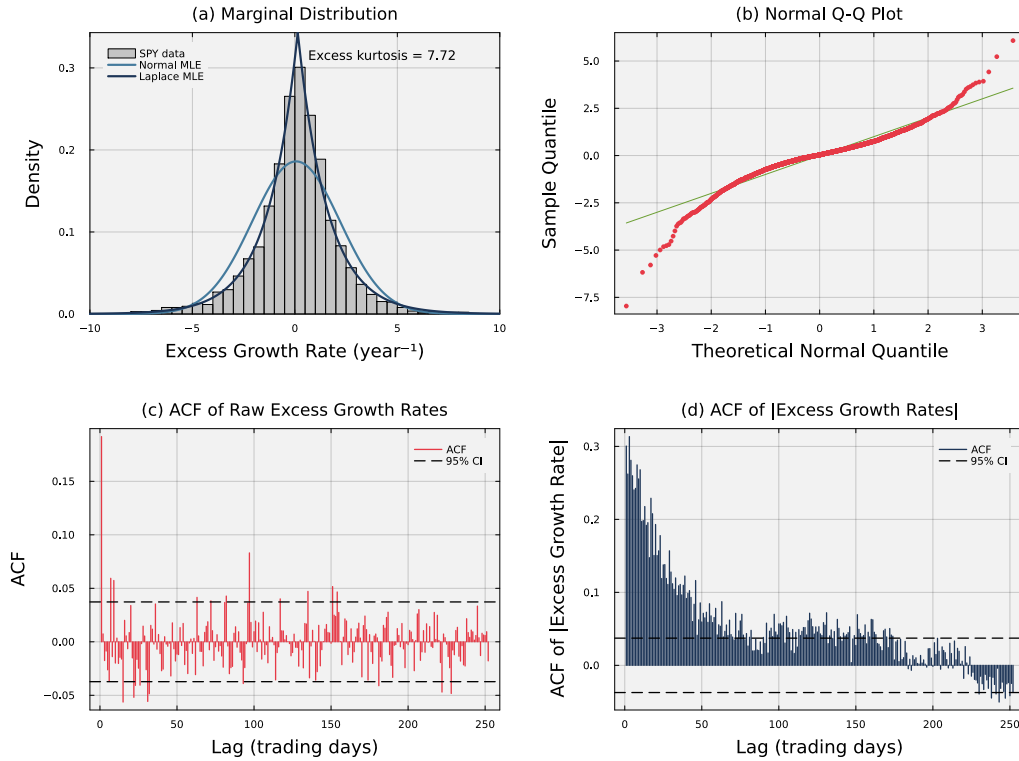


Figure 1: **Empirical stylized facts of SPY daily excess growth rates (2014–2024).** Panel (a) shows the leptokurtic marginal density; the Laplace maximum-likelihood fit tracks the empirical peak and tails while the Gaussian fit does not. Panel (b) shows departure from the diagonal in a normal quantile-quantile plot at both extremes. Panel (c) shows the autocorrelation function of raw excess growth rates, which stays near zero across all lags and is consistent with the efficient markets hypothesis. Panel (d) shows the autocorrelation function of absolute excess growth rates, which decays slowly across the reported lags.

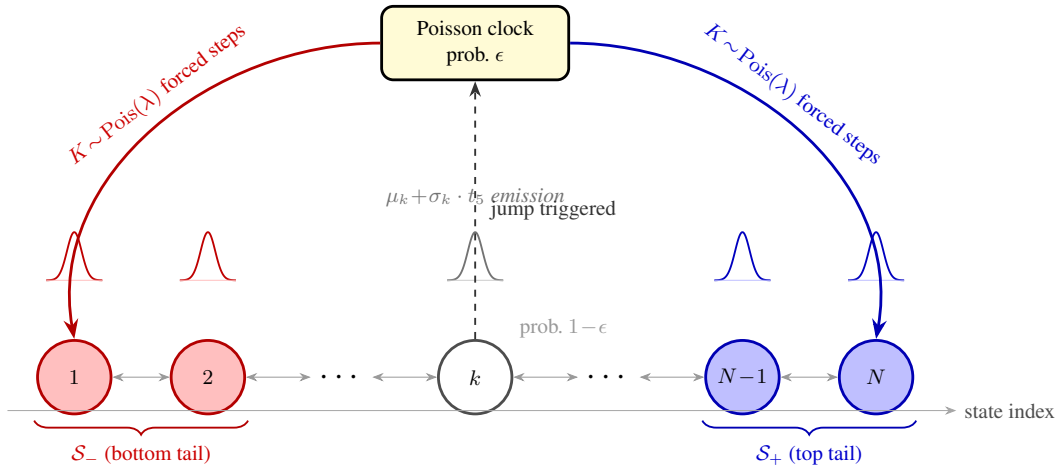


Figure 2: **Architecture of the hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ).** At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1 - \epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ , dashed upward arrow to the Poisson clock). During a jump episode each of the $K \sim \text{Poisson}(\lambda)$ forced steps independently draws the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) with probability p_{neg} or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) otherwise, before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional location-scale Student- t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

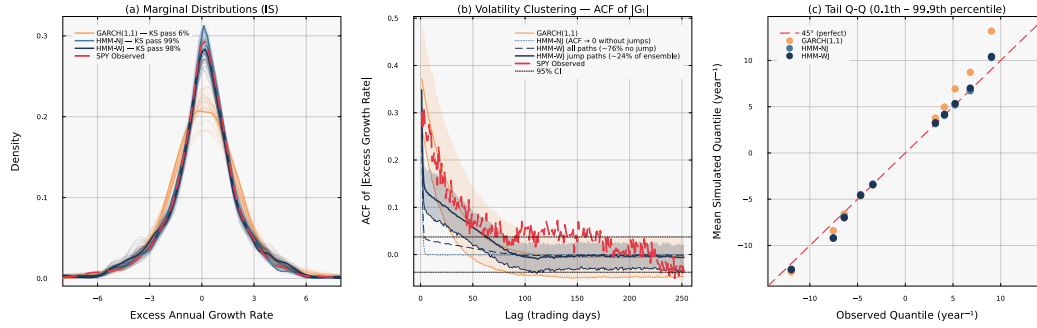


Figure 3: **Head-to-head in-sample model comparison for SPY** ($N = 100, 1,000$ simulated paths). *Panel (a)*: marginal density of excess growth rates with in-sample KS pass rates annotated per generator. *Panel (b)*: autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths) per generator. *Panel (c)*: tail Q-Q plot at the 0.1st–99.9th percentile region per generator.

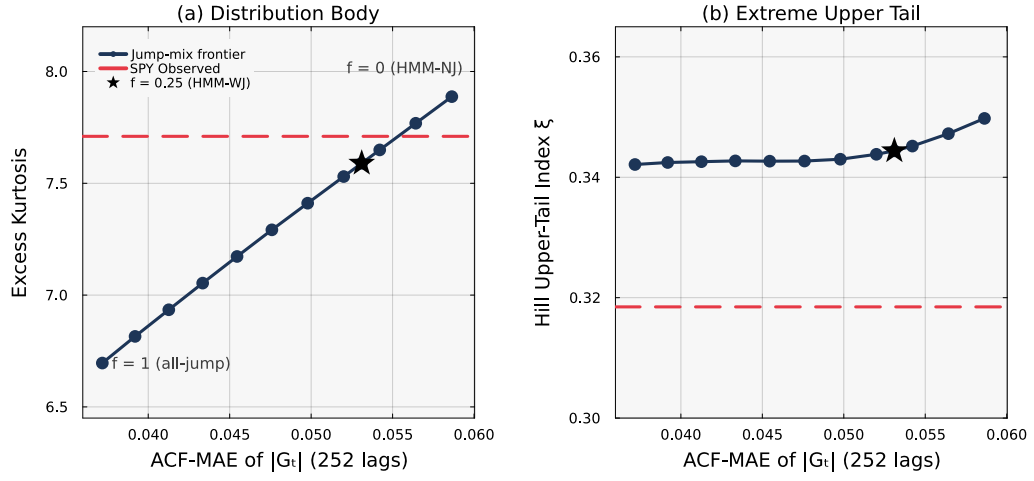


Figure 4: **Jump-mix enrichment frontier for SPY** (in-sample, $N = 100$, pool of 4,000 simulated paths). Resampling the fitted HMM-WJ ensemble to a target jump-path fraction f (the share of accepted paths containing at least one jump) traces the temporal-marginal tradeoff. Both panels share the horizontal axis: absolute-return autocorrelation error (ACF-MAE of $|G_t|$ over lags 1–252); lower values indicate stronger volatility clustering. Panel (a): simulated excess kurtosis. Panel (b): Hill upper-tail index ξ , the raw estimator equal to the reciprocal of the tail exponent. Red dashed lines mark the observed SPY values; the star marks the $f \approx 0.25$ value used throughout. Endpoints $f = 0$ and $f = 1$ correspond to the no-jump generator and an all-jump ensemble.

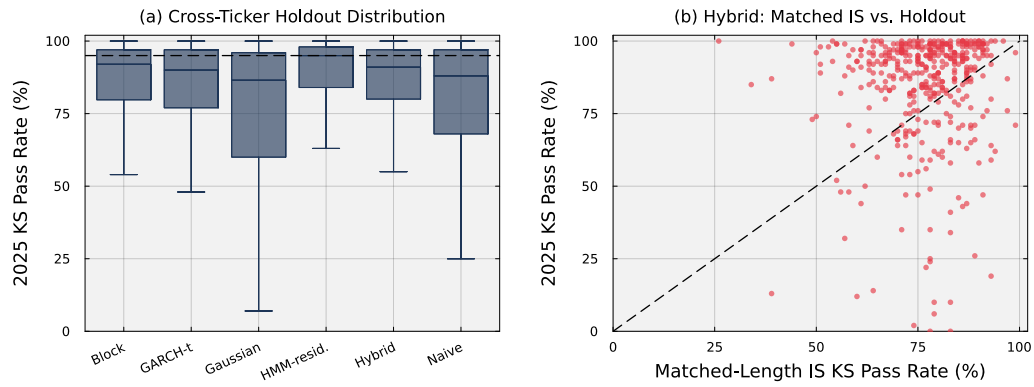


Figure 5: **Cross-ticker distribution of frozen-parameter out-of-sample marginal fit.** Panel (a) shows the distribution of per-ticker 2025 KS pass rates across 100 replications; the dashed line marks 95%. Panel (b) compares each ticker's hybrid KS pass rate against a matched-length training block with its pass rate against the 2025 holdout. The diagonal denotes equal performance.

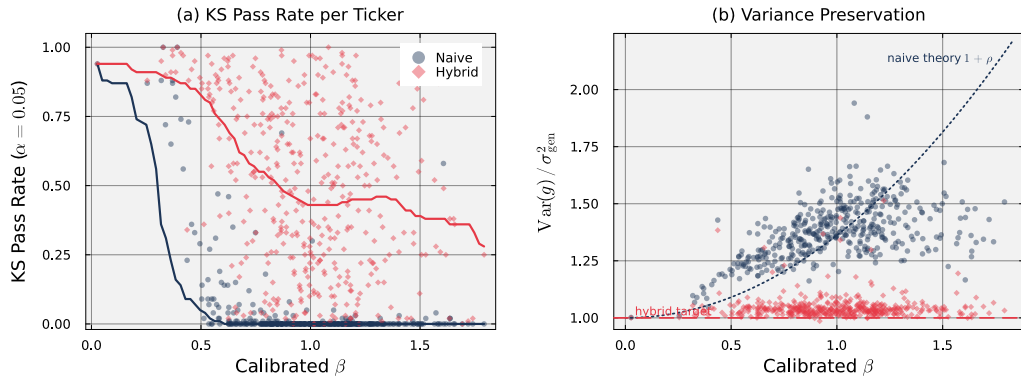


Figure 6: **Hybrid’s variance correction and its KS-pass-rate consequence, per ticker vs calibrated β .** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Left (consequence):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$ for naive composition (blue) and hybrid (orange). *Right (cause):* variance ratio $\text{Var}(g)/\sigma_{\text{gen}}^2$ for the same two methods. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ_{gen}^2 ; the dashed line marks a variance ratio of one.